

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 6 Winter 2025 ( Supply. )**

**Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology**

**Semester : SEMESTER - 6**

**Subject Code & Name: BTAIPE603D - WEB DEVELOPMENT**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which servlet lifecycle method is called only once?  
a) service()    b) init()    c) doGet()    d) destroy()
  - 2 Which WSDL element defines the abstract operations of a web service?  
a) binding    b) service    c) portType    d) message
  - 3 Which Bootstrap class is used to create a responsive fixed-width container?  
a) *.container*    b) *.container-fluid*    c) *.row*    d) *.col-md-12*
  - 4 Which session-tracking method exposes session data in the browser URL?  
a) Cookies    b) Hidden form fields    c) HttpSession    d) URL rewriting
  - 5 Which operator checks both value and type equality in JavaScript?  
a) =    b) ===    c) ==    d) !==
  - 6 *console.log(typeof NaN);* Output?  
a) "number"    b) "NaN"    c) "undefined"    d) "object"
  - 7 Which HTTP response code most accurately represents a successful request where the response body is intentionally empty?  
a) 200    b) 204    c) 302    d) 304
  - 8 SOAP ensures interoperability mainly through:  
a) XML-based messages    b) HTTP dependency  
c) Tight coupling    d) Stateful services
  - 9 Which DOM event phase executes listeners registered with *addEventListener(type, listener, true)*?  
a) Bubbling phase    b) Target phase    c) Capturing phase    d) Default phase
  - 10 PHP file uploads fail despite a valid form. Which setting MOST likely causes this?  
a) *max\_execution\_time*    b) *post\_max\_size*  
c) *memory\_limit*    d) *display\_errors*
  - 11 Node.js achieves high concurrency mainly due to:  
a) Multithreading    b) Blocking I/O    c) Process forking  
d) Event-driven non-blocking I/O

- 12 In WSDL, which element defines service operations independent of transport?  
 a) service      b) binding      c) portType      d) message
- Q2. Solve the following.
- A) State the purpose of HTTP in web communication. Explain how a web client communicates with a web server using HTTP. 6
- B) Write the syntax of an HTML hyperlink and an HTML image tag. Compare HTML and HTML5 elements used for multimedia support. 6
- Q3. Solve the following.
- A) Define JavaScript and list its key features. Explain how JavaScript is executed in a web browser. 6
- B) Write a JavaScript program to find the largest of three numbers. 6
- Q4. Solve Any Two of the following.
- A) Define a Servlet and list its advantages. Explain Servlet architecture with a neat diagram. 6
- B) State the purpose of XML namespaces. Describe in detail the DOM-based XML processing. 6
- C) Write a simple program using a servlet to read user input using request parameters. 6
- Q5. Solve Any Two of the following.
- A) A web-based company wants to create a simple dashboard page that displays the current date and time to users whenever they visit the page. The company uses JavaServer Pages (JSP) for developing its web applications. As a web developer, write a JSP program that displays the current date and time on the webpage. The solution should correctly retrieve the system date and time and format it in a readable way for users. 6
- B) Define JSP and list its features. Explain the JSP execution lifecycle. 6
- C) Differentiate between REST and SOAP web services using any SIX points. 6
- Q6. Solve Any Two of the following.
- A) List any SIX PHP data types with suitable examples and syntax. 6
- B) A web-based company wants to allow users to log in to their portal using a username and password. The company uses PHP for server-side processing and a MySQL database to store user credentials. As a PHP developer, write a PHP program to process a login form. The program consists of the following: 6
- Retrieve the username and password submitted from the form.
  - Validate the credentials against stored values (you can assume predefined variables for simplicity).
  - Display a success message if the login is correct, or an error message if it is incorrect.
- C) Explain how PHP can connect to a MySQL database, including the steps, functions, and methods used to establish the connection, perform queries, and handle results. 6

**\*\*\* End \*\*\***